

App Dev-1: Project Report

1. Student Details

Name: Shubham Verma

Roll Number: 23dp3000062

Email: 23dp3000062@ds.study.iitm.ac.in

2. Project Details

Project Title: Placement Portal Application

Problem Statement: To design and build a web-based application that streamlines the campus placement process by enabling Admin (Institute), Companies, and Students to interact with the system based on their respective roles — allowing companies to post drives, students to apply, and admins to oversee and manage the entire process.

Approach: The app was built using Flask as the backend framework with a modular structure consisting of `app.py`, `config.py`, `models.py`, and `routes.py`. It uses SQLAlchemy ORM with an SQLite database to manage users, placement drives, and applications. Role-based access control is implemented for three user types — Admin, Company, and Student — each with dedicated dashboards and functionality. Companies can create and manage placement drives, students can browse and apply to approved drives, and the admin can approve or reject companies and drives, monitor all activity through a central dashboard.

3. AI/LLM Declaration

AI/LLM Usage: I utilized **ChatGPT** and **Copilot** to assist in designing and refining the SQLAlchemy database schema, including model definitions, relationship mappings, and constraint configurations. A little amount of assistance was also taken for bootstrap styling and structuring Jinja2 templates. The overall extent of AI/LLM usage is approximately 20–25%, primarily concentrated in the database layer and minor template scaffolding. All core business logic, debugging, integration, and final decision-making were done manually and independently.

4. Technologies and Frameworks Used

<u>Technology / Library</u>	<u>Purpose</u>
<u>Flask</u>	<u>Core backend web framework</u>
<u>SQLAlchemy</u>	<u>Object Relational Mapper for SQLite database</u>
<u>Jinja2</u>	<u>Template engine for rendering dynamic HTML pages</u>
<u>Bootstrap 5</u>	<u>Frontend styling and responsive UI design</u>
<u>SQLite</u>	<u>Lightweight local database for storing all portal data</u>
<u>Werkzeug</u>	<u>Password hashing and security utilities</u>
<u>python-dotenv</u>	<u>Managing environment variables via .env file</u>
<u>Flask Sessions</u>	<u>User authentication and role-based session management</u>

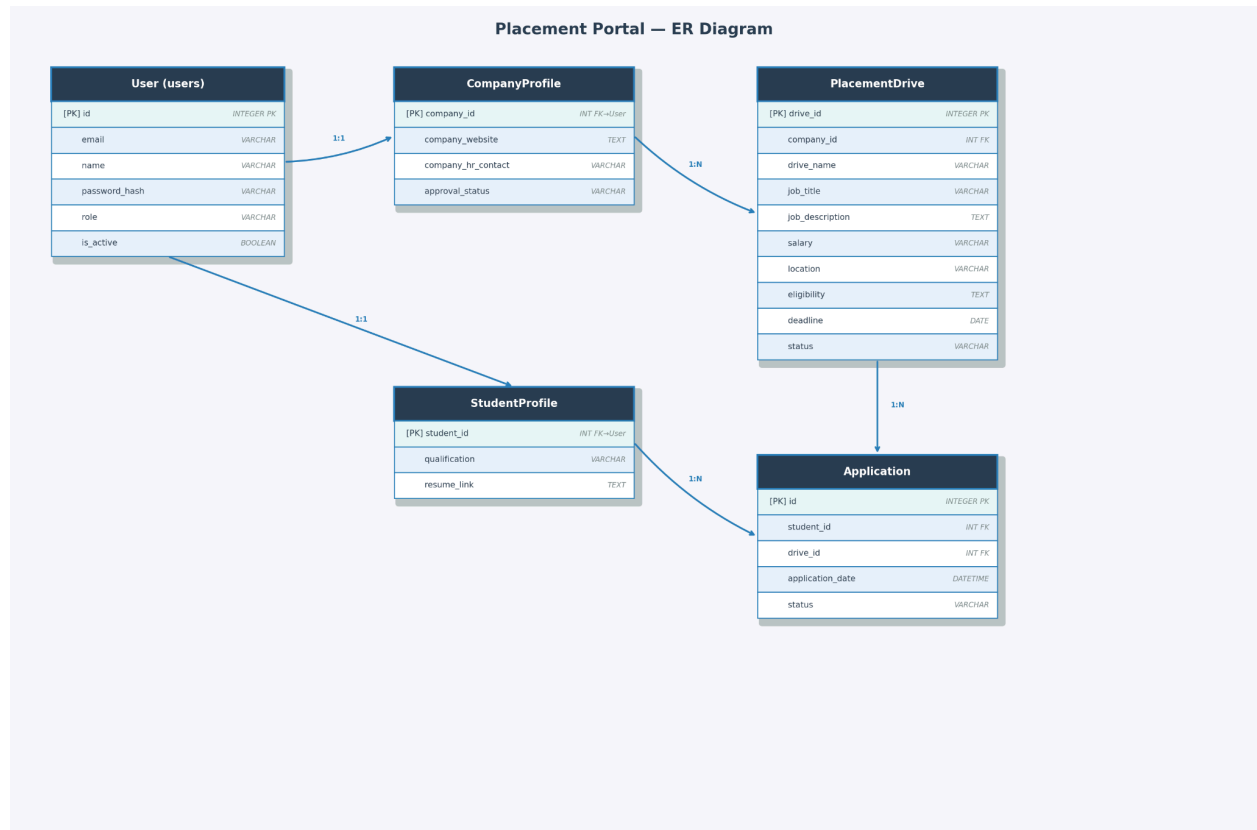
5. Database Schema / ER Diagram

Tables:

1. User — stores account details for all roles (id, email, name, password_hash, role, is_active)
2. CompanyProfile — stores company-specific information (company_id, company_website, company_hr_contact, approval_status)
3. StudentProfile — stores student-specific information (student_id, qualification, resume_link)
4. PlacementDrive — stores placement drive details posted by companies (drive_id, company_id, drive_name, job_title, job_description, salary, location, eligibility, deadline, status)
5. Application — tracks student applications to drives (id, student_id, drive_id, application_date, status)

Relationships:

- One-to-One → User → CompanyProfile
- One-to-One → User → StudentProfile
- One-to-Many → CompanyProfile → PlacementDrive
- One-to-Many → PlacementDrive → Application
- One-to-Many → StudentProfile → Application



8. Video Presentation

Drive Link:

https://drive.google.com/file/d/18yRIJgaKSCDH7-1ECP-TLFUWfGALdsqV/view?usp=drive_sdk